

Quantum-safe cryptography: the quiet migration reshaping digital sovereignty

A companion report to the QSC Atlas, with the governance-credibility evaluations

Companion to the QSC Atlas (qsc-atlas.vercel.app). Research note, updated June 2026. [Author and affiliation]

Abstract. The QSC Atlas maps, country by country, how governments are rebuilding public-key cryptography against the prospect of a quantum computer able to break it. This updated report is concise by design. It sets out the two-axis classification behind the Atlas, a coordination posture and a standards role; the regulatory layer that records what each instrument requires; and, new to this edition, an evaluative layer that scores how credible each national plan is on six criteria. It explains how the evaluations are measured and verified, and reports what they show across the 53 recorded countries: credibility is high where a country has committed to a bloc and thin where it has not, and the weakest link everywhere is execution, the inventories and measurement that turn a plan into practice. The full scores are in the appendix.

1. The instrument and its question

Public-key cryptography secures most of what the digital economy runs on. A sufficiently powerful quantum computer would break the asymmetric algorithms that work depends on; no such machine exists, and the date of its arrival cannot be known. Governments, regulators, standards bodies and technology firms are nonetheless already rebuilding those foundations against the anticipated threat, and the QSC Atlas records that rebuilding as it happens. Two features make it hard to read from any single vantage point: cryptographic protocols are infrastructure in Star's (1999) sense, visible only on breakdown and migrated rather than switched; and no single authority governs the transition, which is coordinated instead across national agencies, regional roadmaps and transnational standards bodies. The Atlas shows that scattered structure rather than assuming a centre that does not exist, and now also asks a further question: not only where a country stands, but how credible its plan is.

2. Theoretical principles

The Atlas rests on the standards-governance literature, which treats technical standards as instruments of power with distributional consequences rather than neutral specifications. Buthe and Mattli (2011) show how first-mover advantage, institutional access and technical capacity determine whose preferences become embedded in global rules; Yates and Murphy (2019) trace the same asymmetry across a century of voluntary consensus standard-setting. The post-quantum case reproduces it: when the United States published its first three standards in 2024 (NIST, 2024), a contested question became a settled technical fact that other jurisdictions adopt or must interoperate with. Bradford's (2020) Brussels Effect is the instructive contrast, because in cryptography the direction reverses: the standards originate in Washington and travel through the global deployment of United States technology firms, while European institutions contextualise rather than make.

Beneath the standards reading sits the governance-of-expectations literature (Borup et al., 2006; Konrad, 2006). Because the threat's timing is genuinely unknowable, coordination proceeds through shared expectations that can harden into commitment while remaining revisable (Budde and Konrad, 2019); the governance stays provisional. The evaluative layer added in this edition takes its criteria from the same register: it adapts the six-criteria comparative method Paglieri et al. (2025) built for national quantum strategies, sharpened by Vance's (2025) split of evaluation into process, impact and cost-benefit, and draws two of its checks from the CEPS governance recommendations (Pupillo et al., 2025). A credibility score is therefore a reading at a date, not a verdict.

3. The two-axis classification

Every recorded country is described on two dimensions, kept apart because one label cannot carry both. The coordination posture answers which collective transition a country belongs to, and takes one of four values: the EU coordinated roadmap, the NIST-led bloc, a sovereign bloc, or engaged but unaligned. The standards role answers its relationship to the algorithms: standard-maker, contextualiser, standard-taker, or sovereign developer. A maker writes the baseline; a contextualiser adopts it but adds national requirements such as mandatory hybridisation or certification; a taker adopts it broadly as published; a sovereign developer designs its own. The posture is the map colour, the role a profile badge. Figure 1 shows why the split matters: the sovereign and engaged groups each sit on a single role, but the committed blocs divide across roles, with the United States the lone maker, seven contextualisers, and the majority takers. A one-colour scheme would hide that.

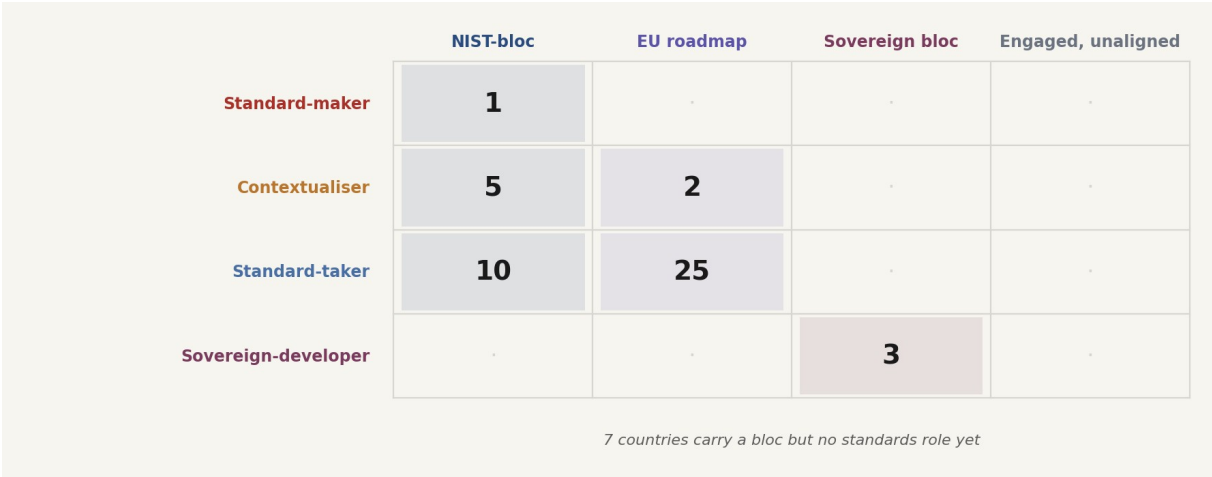


Figure 1. The two-axis classification: coordination bloc by standards role. 53 recorded countries.

4. The regulatory layer

Between the classification and the evaluation sits a regulatory layer that turns a label into an obligation. For each country it records the main governing instrument, its legal status, binding, soft-law or none, and the plain duty it places on regulated entities. The distinction is load-bearing: a recommendation and a statute would otherwise read alike. Of the 53 recorded countries, 39 carry at least one binding

instrument, 6 rest on soft law alone, and 8 have no governing instrument on record. Binding EU duties under NIS2 and DORA read differently from a soft national roadmap, which reads differently again from nothing at all, and the evaluative layer draws directly on these fields.

5. Measuring credibility: the evaluative layer

The evaluative layer asks not whether a plan exists but how credible it is. Six criteria, relevance, coherence, effectiveness, efficiency, governance quality, and impact, become eleven sub-criteria, each scored 0 (absent), 1 (partial) or 2 (present) against a stated description, and reported as a six-axis profile rather than a single number (Figure 2). Relevance reads the transition plan and the threat articulation; coherence, the integration with wider policy and the breadth of consultation; effectiveness, the milestones, the inventory duty and any measurement framework; efficiency, whether the instrument's strength matches the objective; governance quality, the mandate and the skills base; and impact, the exposure-reducing adoption of the baseline. Each sub-score is derived from a field the Atlas already records, so it is traceable: milestones and target years drive relevance and effectiveness, the legal status drives efficiency, the recorded instruments and obligations drive coherence and the inventory check, and the standards role with the hybrid stance drives impact.

Three rules keep the scores defensible. No sub-score is entered unless it traces to a sourced instrument; an axis with no supporting field reads as not assessed, never a default, and is the mean of the sub-criteria that could be assessed. A second reviewer scores each country independently and the two reconcile before a profile is shown. The Atlas confidence grade is carried onto each axis, so a country assessed on thin evidence reads as tentative. The scores in this report are a first pass, derived mechanically from the recorded fields by these rules, with reviewer reconciliation still under way.

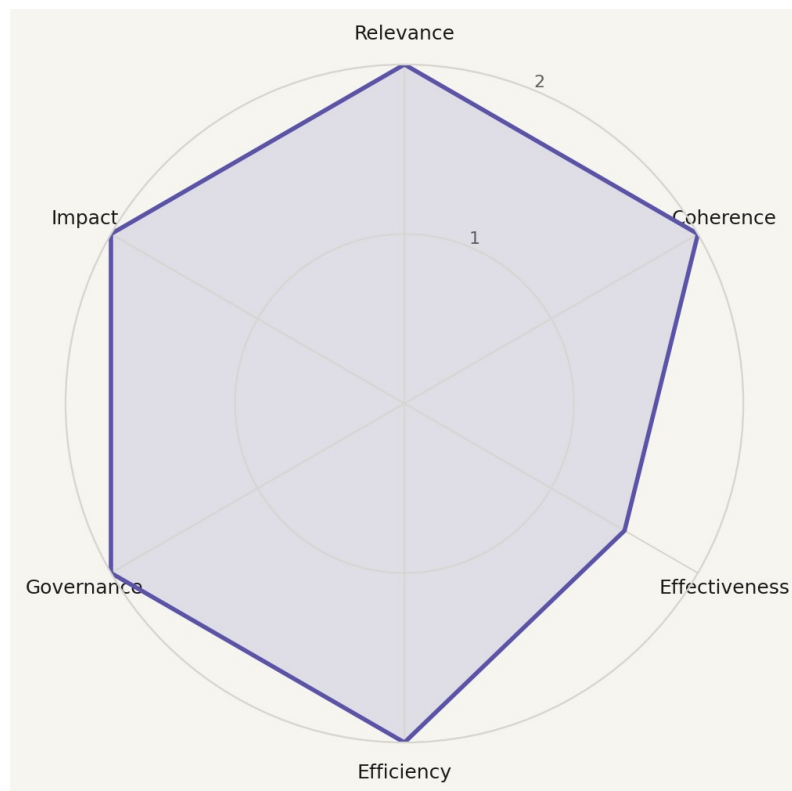


Figure 2. How a country reads: the six-axis credibility profile for France, an EU contextualiser. High on relevance, coherence, efficiency, governance and impact; a step lower on effectiveness, where the milestones are not-earlier-than windows and no dedicated measurement duty is recorded.

6. What the evaluations show

Read across the six axes, the scores have a consistent shape (Figure 3). Relevance and impact are highest, at 1.70 and 1.87, because plans exist and the NIST baseline is adopted almost everywhere. Effectiveness is the lowest axis, at 1.31, because the fields it reads, mandated inventories and measurement frameworks, are the thinnest in the record. The finding is that plans outrun the machinery to execute them: the transition is widely declared and widely under-instrumented.

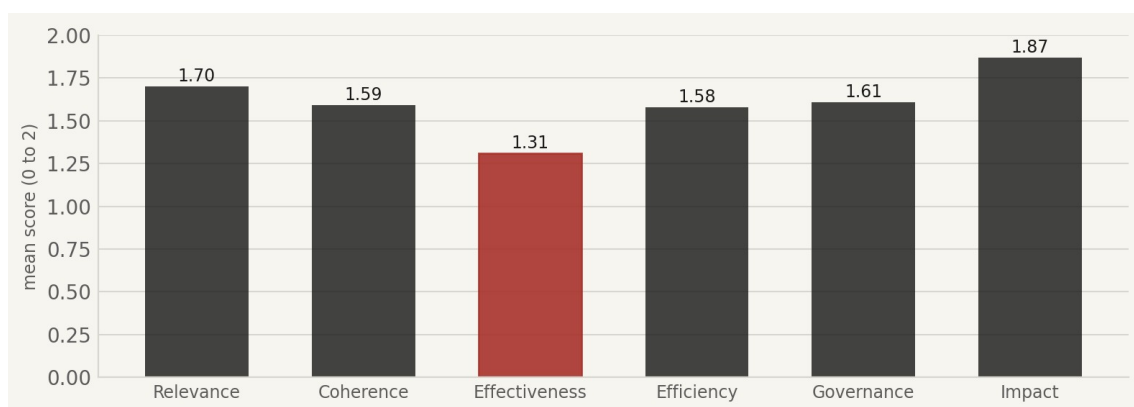


Figure 3. Mean credibility by axis across the recorded countries. Effectiveness, the execution machinery of inventories and measurement, is the weakest link everywhere.

Read by coordination bloc, credibility concentrates in the committed (Figure 4). The NIST-led and EU blocs score almost identically, 1.81 and 1.79, confirming that on credibility, as on algorithms, the two committed blocs behave alike. The sovereign bloc sits lower at 1.33, its independent track less legible in the fields the layer reads. The engaged-but-unaligned group scores far below the rest, at 0.58: the credibility gap is concentrated among the states that have named no bloc, standard or timeline. This is the classification and the evaluation agreeing, commitment and credibility travel together, and it is the clearest single result of the layer.

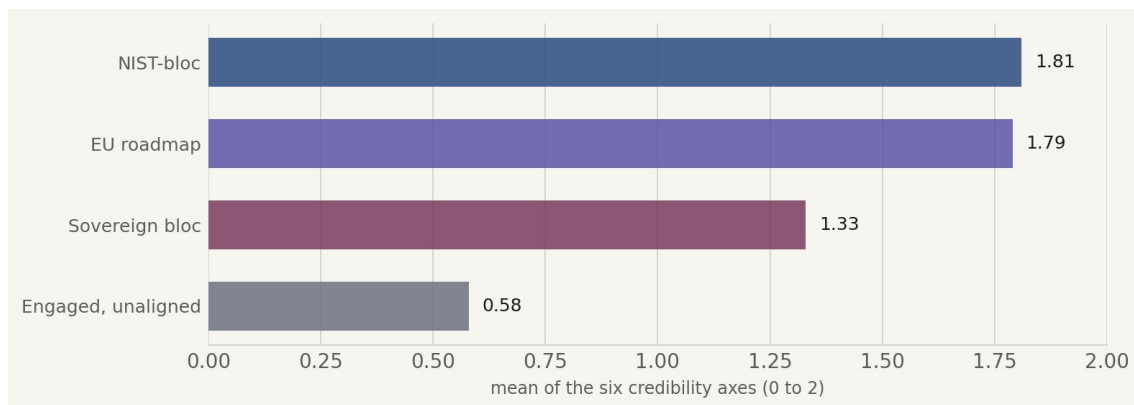


Figure 4. Mean credibility by coordination bloc. Commitment and credibility travel together; the gap is concentrated among the unaligned.

7. Why it matters

Theoretically, the case extends standards-governance and governance-of-expectations accounts into a transnational, multi-level setting: expectations generated within one national system acquire binding force elsewhere through market mechanisms and institutional referencing rather than formal agreement (Buthe and Mattli, 2011; Carlsson, 2006). Methodologically, separating coordination from role avoids a bloc illusion in which the EU reads as a rival standards camp when on the algorithm axis it runs the same baseline under a regulatory wrapper; and scoring credibility on sourced fields, with confidence carried and reconciliation required, keeps the reading defensible rather than asserted. Geopolitically, whoever makes the baseline holds structural power over the terms on which everyone else secures their systems, which is why the single-maker finding is the central fact of the field; the credibility gap among the unaligned marks where the contest for the undecided middle will be decided.

8. Conclusion

The QSC Atlas is a living instrument for a transition that has not finished and may not finish on any predictable schedule. Its contribution is to make a leaderless migration legible: which collective each country coordinates with, whose algorithms it runs, what binds it, and now how credible its plan is. The evaluations are provisional by construction and move as the evidence does. Two things are already clear: credibility follows commitment, and execution is the common weak point. The appendix records the current classification and the credibility scores for all 53 recorded countries.

Appendix A. Classification of the recorded countries

Coordination bloc, standards role and legal status of the governing instrument, from the QSC Atlas.

Source: institutional documents on each country profile.

Country	Bloc	Role	Legal
Australia	NIST-bloc	Contextualiser	Binding
Austria	EU roadmap	Standard-taker	Binding
Bahrain	Engaged, unaligned	Not yet declared	None
Belgium	EU roadmap	Standard-taker	Binding
Brazil	NIST-bloc	Standard-taker	Binding
Brunei	Engaged, unaligned	Not yet declared	None
Bulgaria	EU roadmap	Standard-taker	Binding
Canada	NIST-bloc	Standard-taker	Binding
Chile	Engaged, unaligned	Not yet declared	None
China	Sovereign	Sovereign-developer	Soft law
Croatia	EU roadmap	Standard-taker	Binding
Cyprus	EU roadmap	Standard-taker	Binding
Czechia	EU roadmap	Standard-taker	Binding
Denmark	EU roadmap	Standard-taker	Binding
Djibouti	Engaged, unaligned	Not yet declared	None
Estonia	EU roadmap	Standard-taker	Binding
Finland	EU roadmap	Standard-taker	Binding
France	EU roadmap	Contextualiser	Binding
Germany	EU roadmap	Contextualiser	Binding
Greece	EU roadmap	Standard-taker	Binding
Hungary	EU roadmap	Standard-taker	Binding
India	NIST-bloc	Contextualiser	Binding
Indonesia	Engaged, unaligned	Not yet declared	None
Ireland	EU roadmap	Standard-taker	Binding
Israel	NIST-bloc	Standard-taker	Binding
Italy	EU roadmap	Standard-taker	Binding
Japan	NIST-bloc	Contextualiser	Binding

Country	Bloc	Role	Legal
Latvia	EU roadmap	Standard-taker	Binding
Lithuania	EU roadmap	Standard-taker	Binding
Luxembourg	EU roadmap	Standard-taker	Binding
Malaysia	NIST-bloc	Standard-taker	Soft law
Malta	EU roadmap	Standard-taker	Binding
Netherlands	EU roadmap	Standard-taker	Soft law
New Zealand	NIST-bloc	Standard-taker	Binding
Norway	NIST-bloc	Standard-taker	Binding
Poland	EU roadmap	Standard-taker	Binding
Portugal	EU roadmap	Standard-taker	Binding
Romania	EU roadmap	Standard-taker	Binding
Russia	Sovereign	Sovereign-developer	Binding
Saudi Arabia	NIST-bloc	Standard-taker	Binding
Singapore	NIST-bloc	Contextualiser	Soft law
Slovakia	EU roadmap	Standard-taker	Binding
Slovenia	EU roadmap	Standard-taker	Binding
South Korea	NIST-bloc	Contextualiser	Binding
Spain	EU roadmap	Standard-taker	Binding
Sweden	EU roadmap	Standard-taker	Binding
Thailand	NIST-bloc	Standard-taker	Soft law
Türkiye	Engaged, unaligned	Not yet declared	None
Ukraine	Engaged, unaligned	Not yet declared	None
United Arab Emirates	NIST-bloc	Standard-taker	Binding
United Kingdom	NIST-bloc	Standard-taker	Soft law
United States	NIST-bloc	Standard-maker	Binding
Vietnam	Sovereign	Sovereign-developer	None

Appendix B. Governance-credibility scores

Six-axis credibility scores (0 to 2 per axis; a dash means the axis could not be assessed from the recorded fields). First pass, derived from the recorded fields; reviewer reconciliation under way. Rel relevance, Coh coherence, Eff effectiveness, Effi efficiency, Gov governance quality, Imp impact.

Country	Rel	Coh	Eff	Effi	Gov	Imp	Conf.
Australia	2.0	2.0	1.5	2.0	2.0	2.0	Medium
Austria	2.0	1.5	1.5	2.0	0.0	2.0	High
Bahrain	0.5	0.5	0.0	0.0	1.0	1.0	Medium
Belgium	2.0	2.0	1.5	2.0	2.0	2.0	High
Brazil	2.0	2.0	1.5	2.0	2.0	2.0	Medium
Brunei	0.5	0.5	0.0	0.0	1.0	1.0	Low
Bulgaria	2.0	1.5	1.5	2.0	1.0	2.0	High
Canada	2.0	2.0	1.5	2.0	2.0	2.0	High
Chile	0.5	0.5	0.0	0.0	2.0	1.0	Medium
China	1.0	1.5	0.5	1.0	1.5	2.0	High
Croatia	2.0	1.5	1.5	2.0	1.0	2.0	High
Cyprus	2.0	1.5	1.5	2.0	1.0	2.0	High
Czechia	2.0	2.0	1.5	2.0	2.0	2.0	High
Denmark	2.0	2.0	1.5	2.0	2.0	2.0	High
Djibouti	0.5	0.5	0.0	0.0	1.0	1.0	Low
Estonia	2.0	1.5	2.0	2.0	0.0	2.0	High
Finland	2.0	2.0	2.0	2.0	2.0	2.0	High
France	2.0	2.0	1.5	2.0	2.0	2.0	High
Germany	2.0	2.0	1.5	2.0	2.0	2.0	High
Greece	2.0	1.5	1.5	2.0	2.0	2.0	High
Hungary	2.0	2.0	1.5	2.0	2.0	2.0	High
India	2.0	2.0	2.0	2.0	2.0	2.0	Medium
Indonesia	0.5	0.5	0.0	0.0	2.0	1.0	Medium
Ireland	2.0	1.5	1.5	2.0	1.0	2.0	High
Israel	2.0	2.0	2.0	2.0	2.0	2.0	Medium
Italy	2.0	2.0	1.5	2.0	2.0	2.0	High
Japan	2.0	2.0	1.5	2.0	2.0	2.0	Medium
Latvia	2.0	1.5	1.5	2.0	1.0	2.0	High
Lithuania	2.0	2.0	2.0	2.0	2.0	2.0	High
Luxembourg	2.0	2.0	2.0	2.0	2.0	2.0	High
Malaysia	1.0	1.5	1.0	1.0	2.0	2.0	High
Malta	2.0	1.5	1.5	2.0	0.0	2.0	High
Netherlands	1.5	2.0	1.0	1.0	2.0	2.0	High
New Zealand	2.0	1.5	2.0	2.0	2.0	2.0	Medium
Norway	2.0	2.0	2.0	2.0	2.0	2.0	Medium
Poland	2.0	2.0	1.5	2.0	2.0	2.0	High
Portugal	2.0	1.5	1.5	2.0	1.0	2.0	High
Romania	2.0	2.0	1.5	2.0	2.0	2.0	High
Russia	1.5	1.5	0.5	2.0	1.0	2.0	High
Saudi Arabia	1.5	1.5	0.5	2.0	2.0	2.0	Medium
Singapore	1.5	2.0	2.0	1.0	2.0	2.0	Medium
Slovakia	2.0	1.5	1.5	2.0	0.0	2.0	High
Slovenia	2.0	1.5	1.5	2.0	1.5	2.0	High
South Korea	2.0	1.5	1.5	2.0	2.0	2.0	Medium
Spain	2.0	2.0	2.0	2.0	2.0	2.0	High
Sweden	2.0	2.0	1.5	2.0	2.0	2.0	High
Thailand	1.5	1.0	1.0	1.0	1.0	2.0	Medium
Türkiye	0.5	0.5	0.0	0.0	2.0	1.0	Low
Ukraine	0.5	0.5	0.0	0.0	1.5	1.0	Low
United Arab Emirates	2.0	2.0	2.0	2.0	2.0	2.0	Medium
United Kingdom	1.5	1.5	1.0	1.0	2.0	2.0	High
United States	2.0	2.0	2.0	2.0	2.0	2.0	High
Vietnam	1.5	1.5	1.0	0.0	2.0	2.0	Low

Axis means across the 53 countries: relevance 1.70, coherence 1.59, effectiveness 1.31, efficiency 1.58, governance 1.61, impact 1.87.

References

- Borup, M., Brown, N., Konrad, K., & van Lente, H. (2006). The sociology of expectations in science and technology. *Technology Analysis & Strategic Management*, 18(3-4), 285-298.
- Bradford, A. (2020). *The Brussels effect: How the European Union rules the world*. Oxford University Press.
- Budde, B., & Konrad, K. (2019). Tentative governing of fuel cell innovation in a dynamic network of expectations. *Research Policy*, 48(5), 1098-1112.
- Buthe, T., & Mattli, W. (2011). *The new global rulers: The privatization of regulation in the world economy*. Princeton University Press.
- Carlsson, B. (2006). Internationalization of innovation systems: A survey of the literature. *Research Policy*, 35(1), 56-67.
- European Parliament and Council. (2022). Directive (EU) 2022/2555 (NIS2); Regulation (EU) 2022/2554 (DORA).
- Konrad, K. (2006). The social dynamics of expectations. *Technology Analysis & Strategic Management*, 18(3-4), 429-444.
- NIST. (2024). FIPS 203, FIPS 204 and FIPS 205: Post-quantum cryptography standards. National Institute of Standards and Technology.
- Paglieri, L., Bonomi Savignon, A., Scalabrini, F., & Costumato, L. (2025). Navigating the quantum frontier: Examining government strategy to the next technological revolution. *Transforming Government: People, Process and Policy*, 19(4).
- Pupillo, L., et al. (2025). Strengthening the EU transition to a quantum-safe world (Task Force Report). Centre for European Policy Studies.
- Star, S. L. (1999). The ethnography of infrastructure. *American Behavioral Scientist*, 43(3), 377-391.
- Vance, A. S. (2025). *Cybersecurity and quantum computing: A quantitative analysis proposing a framework for assessing quantum cybersecurity maturity* [Doctoral dissertation].
- Yates, J., & Murphy, C. N. (2019). *Engineering rules: Global standard setting since 1880*. Johns Hopkins University Press.